

Motor Control

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Learning Outcomes

By the end of this lecture and further reading, students should be able to:

- Describe properties of human movement
- Define motor control and motor learning
- Explain how movement is a function of the individual, task and environment
- Appreciate how an individual's action, perceptual and cognitive systems may impact human movement
- Explain Gentiles taxonomy of motor skill
- Theories of motor control

What do skilled and unskilled movement look like?





What does human movement look like?

Unskilled movements (learners)

- Energy-inefficient, effortful
- Slow (low velocity; low frequency)
- Jerky

- Variable, inconsistent
- Unsuccessful, inaccurate,
- Error-prone
- High attentional demand-controlled
- Reactive

- Non-adaptable, inflexible
- Unable to attend to other tasks
- Poorly coordinated

Skilled movement

- Energy-efficient, effortless
- Fast (high frequency)
- Smooth
- Reproducible, consistent
- Successful, accurate,

- Error-free
- Low attentional demand -automatic
- Anticipatory/predictive
- Adaptable, flexible
- Able to perform tasks simultaneously (dual tasking)
- Well coordinated

What is motor control

“the ability to regulate or direct the mechanisms essential to movement”

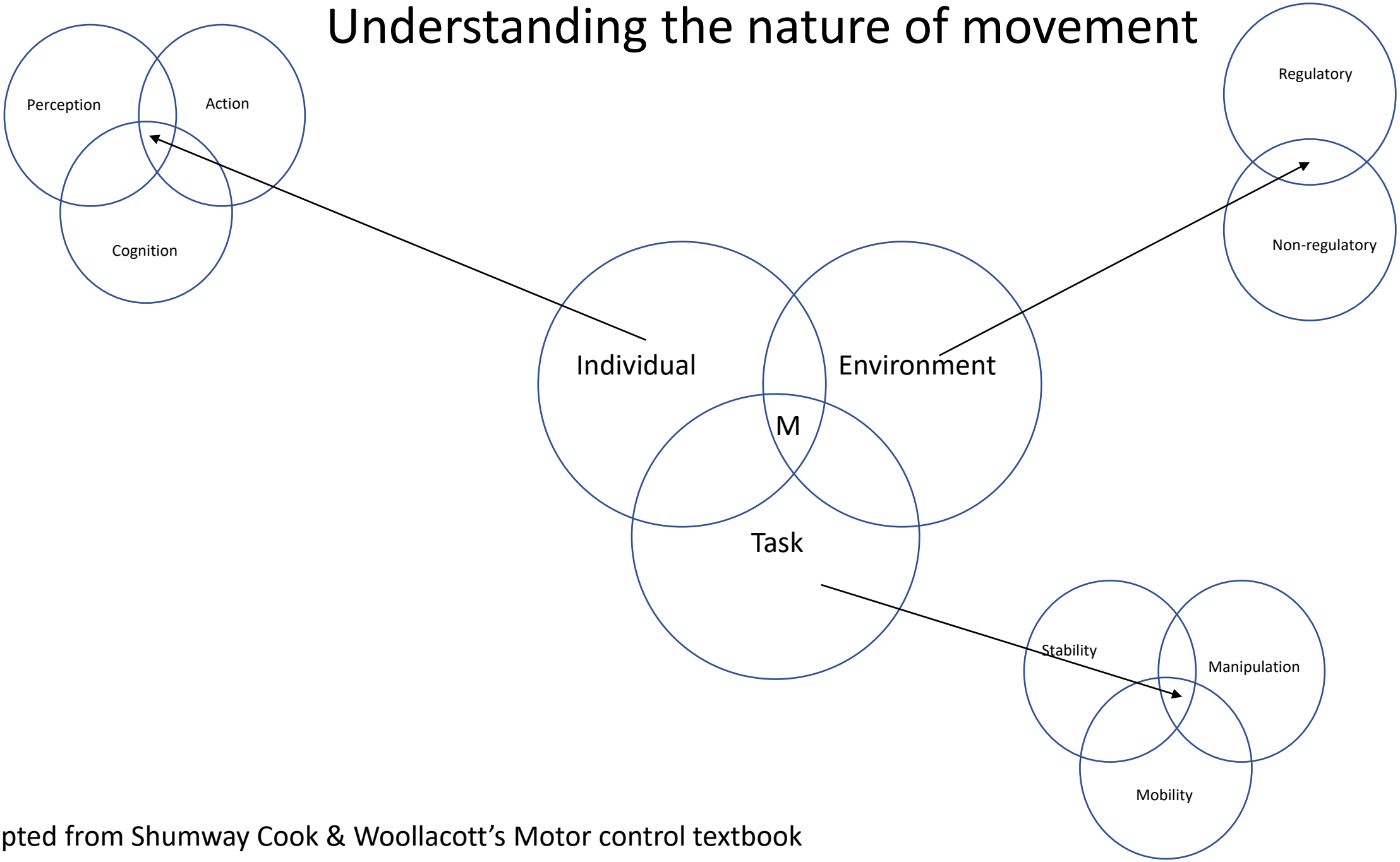
- How the central nervous system (CNS) deals with controlling the many different muscles and joints in different combinations to perform movements
- How sensory information from environment is used to select and control movement both the initiation and modification
- How do our perception of ourselves, the tasks we perform, and the environment in which we are moving influence our movement behaviors
- **Why should therapists study motor control?**

Motor Learning

“field of motor learning is the study of the acquisition and /or modification of movement”

- Motor learning occurs through a process of **experience**, **rehearsal**, **maintenance of learnt movement** and **carry over** of learning to other tasks
    - Experience is the **SENSORY FEEDBACK** to the central nervous system during movements
    - Rehearsal is the repetition of movements that we call **PRACTICE**
    - Learning is the retention of learned skills and can be evaluated in a **RETENTION TEST**
    - Carry over of learning is the ability to utilise elements of learned skills when performing other similar skills and can be evaluated in a **TRANSFER TEST**

Understanding the nature of movement



Adapted from Shumway Cook & Woollacott's Motor control textbook

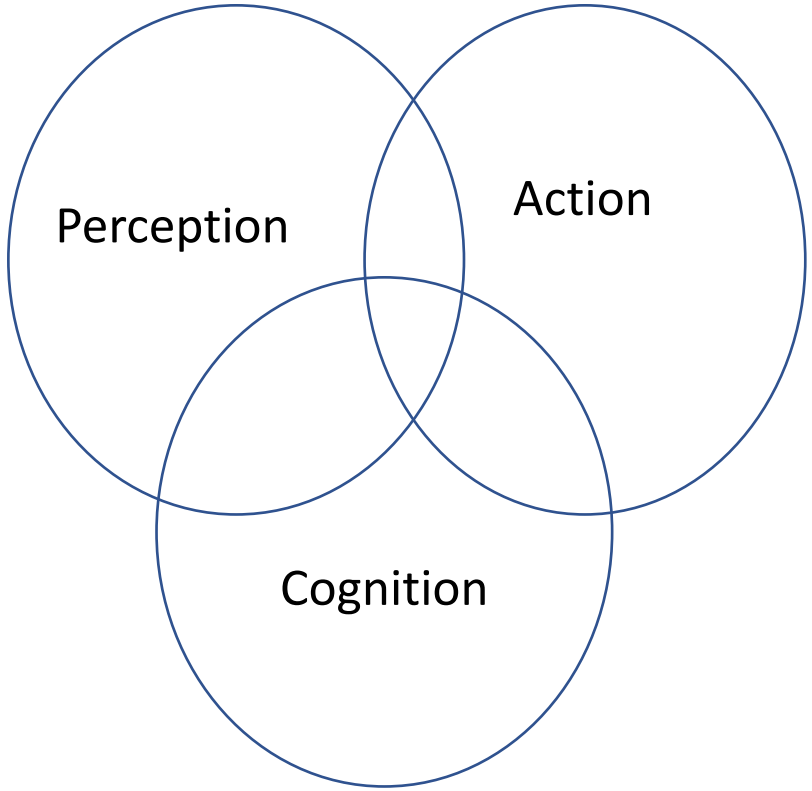
Individual systems underlying motor control: Action/Motor systems

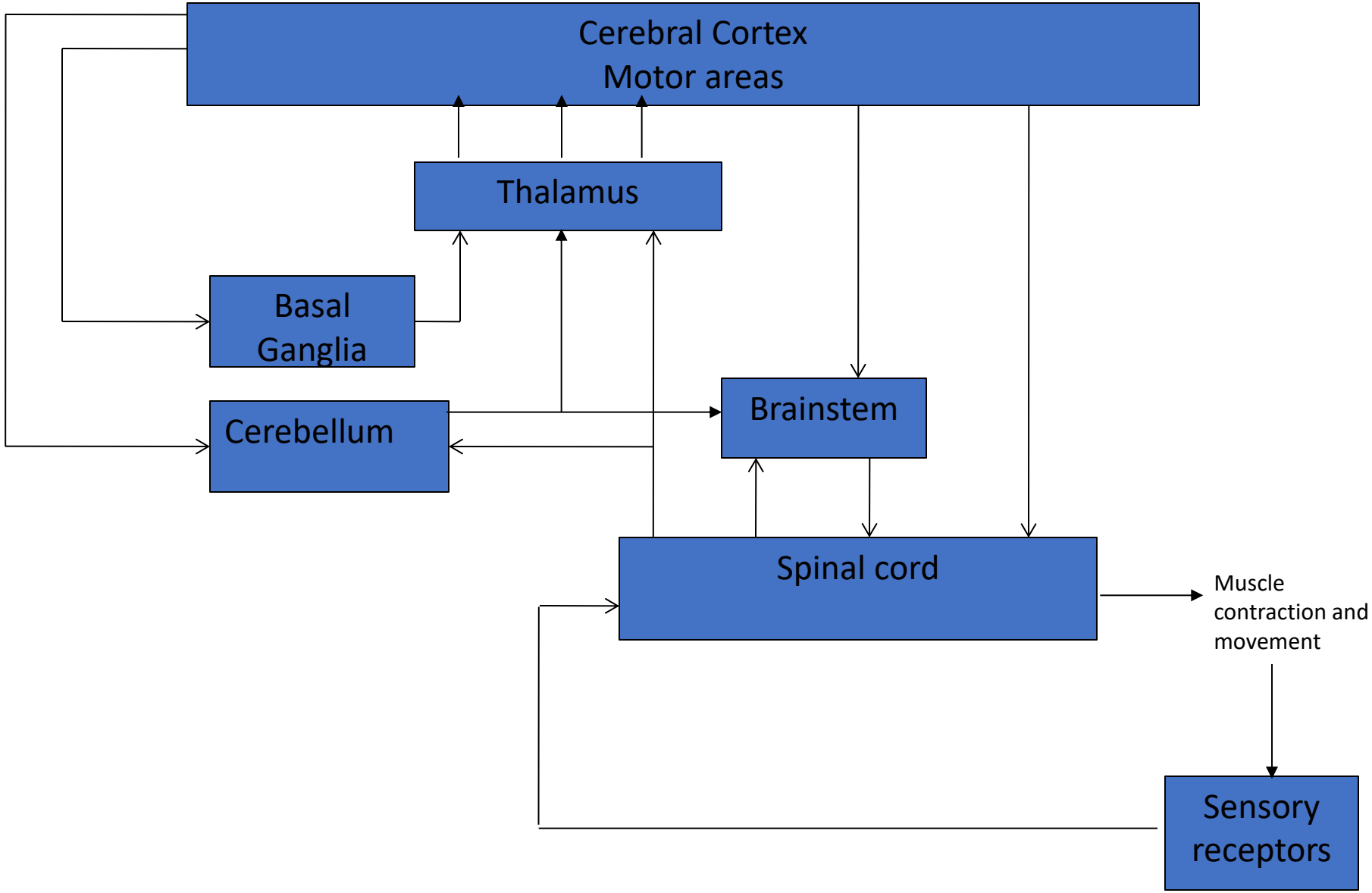
The control of motor output from nervous system to the body's effector systems or muscles

Multiple muscles and joints – controlled during the execution of co-ordinated, functional movement

Multiple ways to do this - multiple equivalent solutions

The problem of choosing among equivalent solutions has been referred to as **degrees of freedom problem** (Bernstein 1967)

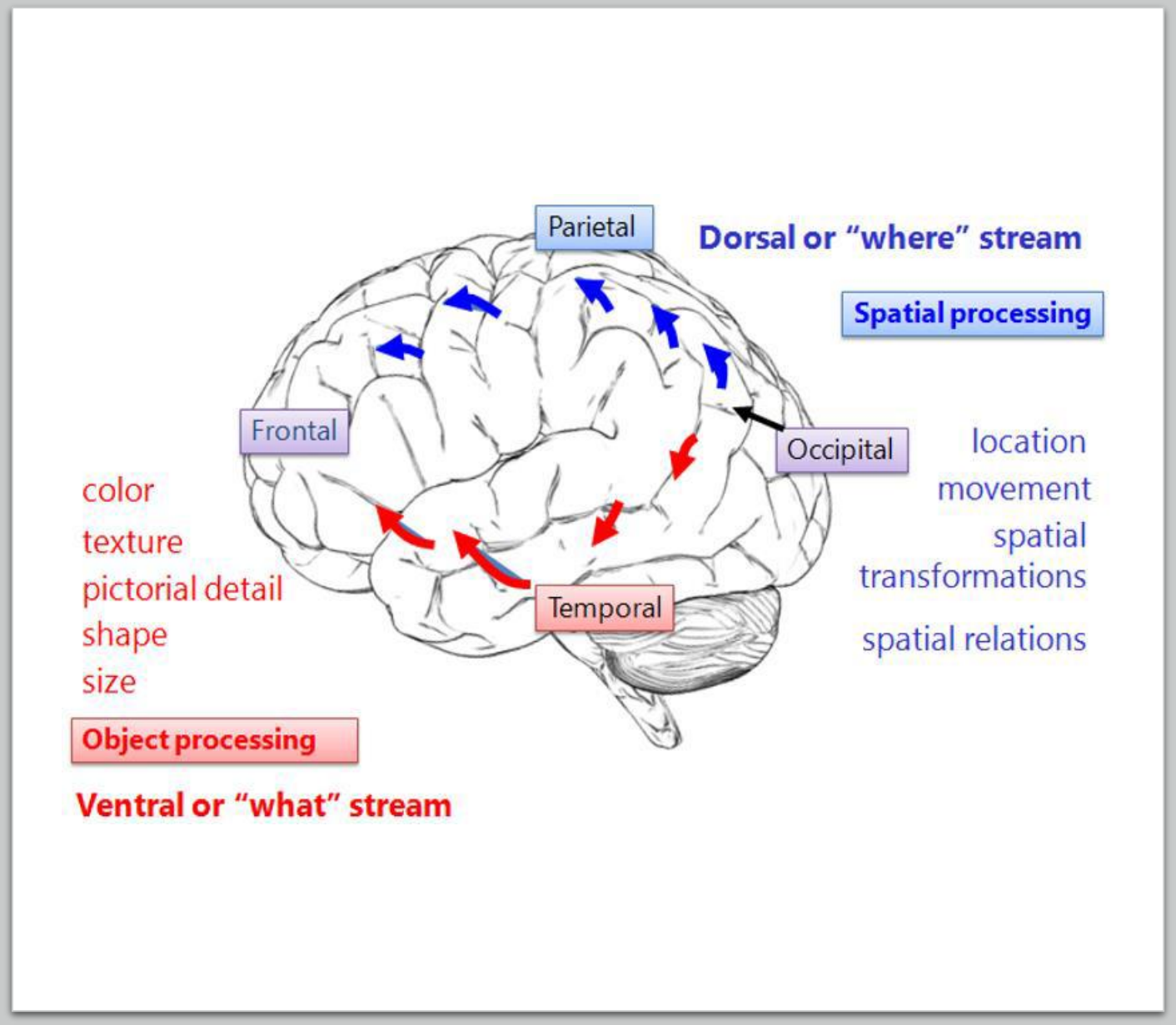




Distributed /network system with parallel processing

Sensory/ Perception

- Perception is the integration of sensory information into meaningful psychological information
- Includes both peripheral sensory mechanisms and higher processing that adds interpretation and meaning to incoming stimuli
- Information about state of the body and the features of the external environment
- It is integral to the ability to act effectively within an environment



Ventral/What

Dorsal/Where

Primary Visual cortex to temporal cortex

Vision for perception

- Characteristics of objects, relations, leading to long-term perceptual representations to permit recognition and identification of objects
- Damage results in intact visually guided grasping movements BUT
  - impaired visual perception
  - difficulty in recognising and identifying people, faces, objects
  - difficulty with orientation/routes e.g., around house and outside

- Primary visual cortex to dorsal parietal cortex
- Vision for action
- Provides information about current location, size and orientation of objects to permit accurate visually guided movements
- Damage results in intact visual perception of objects BUT
  - difficulty handling the complexity of a visual scene
  - difficulty with visual guidance for movement

Cognition, emotion and motivation

- Attention
  - Arousal
  - Focused
  - Divided
- Learning
- Memory
- Problem solving
- Motivation
- Intelligence





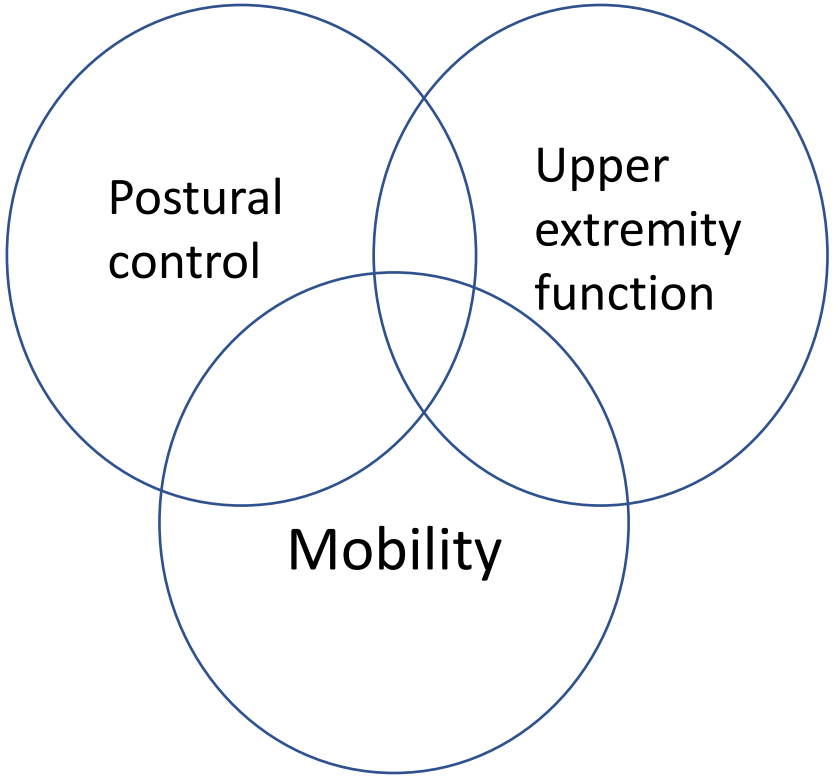


Task constraint of movement

Task can impose constraints on the neural organisation of movement

The nature of the task being performed in part determines the neural organisation of movement

Understanding motor control requires an awareness of how task regulate neural mechanisms controlling movement

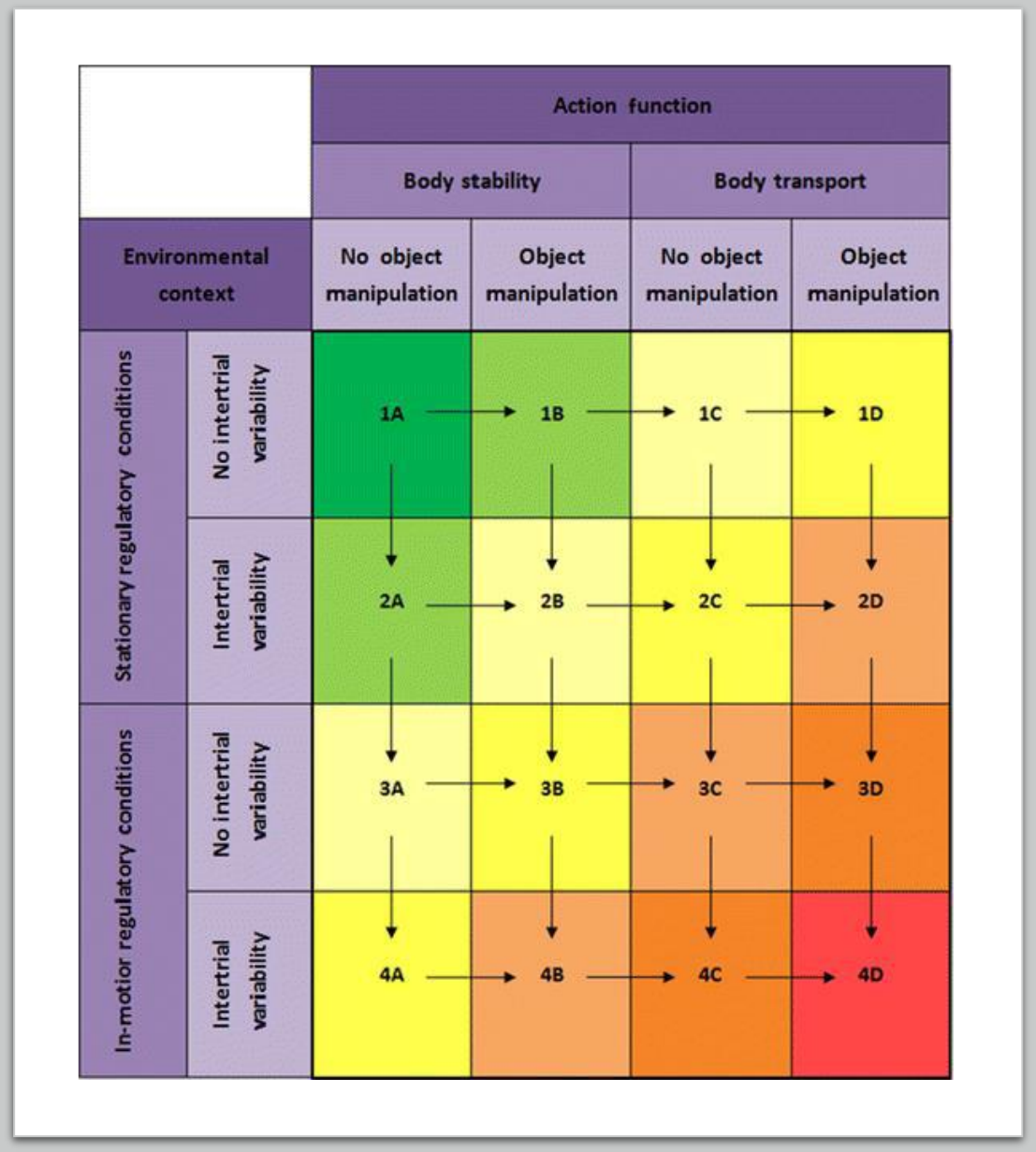


Open and Closed movement tasks

Gentile's taxonomy of motor skill

- A 16-item taxonomy with two key dimensions affecting task complexity.
- The two dimensions are:
- **Environmental Context** (stationary or in-motion regulatory conditions and the degree of inter-trial variability) and
- **Action Function** (body stability or transport and with or without object manipulation)

- Once task complexity is established then this can be used to match and extend the individuals capacity in rehabilitation.



Gentiles' Taxonomy

	Action Function			
	Body Stability		Body Movement	
Environmental Context ↓	No object manipulation	Object manipulation	No object manipulation	Object manipulation
Stationary regulatory condition and no-intertrial variability	1a Practicing a golf swing without a club	1b Hitting balls at a driving range, from the same spot	1c Practicing the same dance steps	1d Practicing the same dance steps with a partner
Stationary regulatory condition and intertrial variability	2a Practicing separate tai chi moves standing on the same spot	2b Hitting ball with cue from different locations on a billiards table	2c Practicing different dance steps on own	2d Practicing different dance steps with a partner
In-motion regulatory condition and no-intertrial variability	3a Riding on a stationary exercise bike at a constant speed	3b Fishing while sitting on a boat and only casting once	3c Practicing lawn bowls delivery with a walking step without a ball	3d Returning a tennis serve from a tennis ball pitching machine
In-motion regulatory condition and intertrial variability	4a Riding on a stationary exercise bike at different speeds	4b Fishing while standing on a boat; throwing cast out in a new spot each attempt	4c Walking in a busy shopping centre	4d Riding a bike on a bike path

Environmental factors

Regulatory





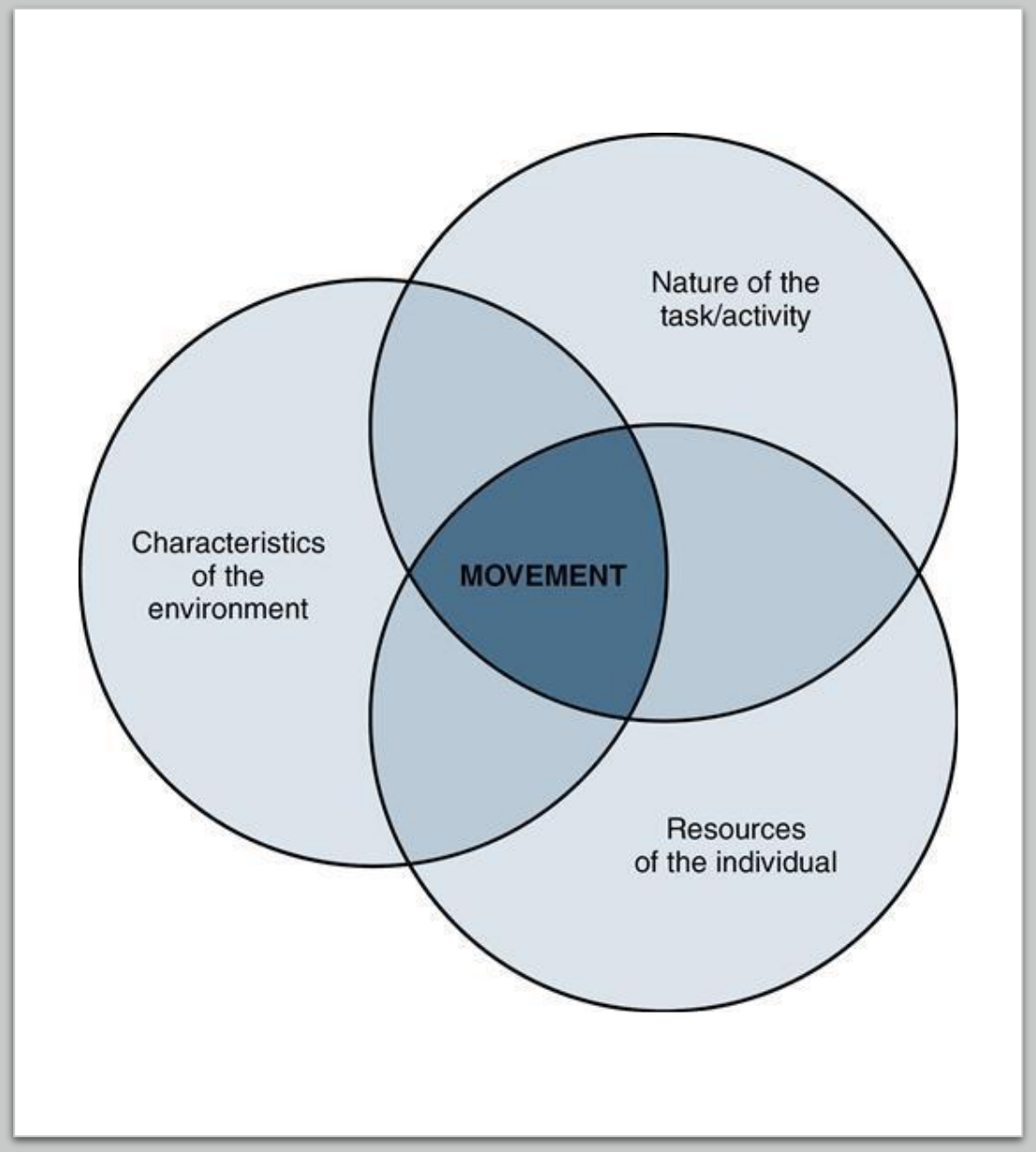


Non-regulatory

- Facilitators
  - Comfortable temperature, calm, good lighting
- Distractors
  - Too hot/too cold, interruptions, noise, wind, choppy sea

Recap

- **Human movement is the product of an interaction between three components – Task, Individual and Environment**–Shumway Cook and Woollacott
- “movement emerges from the interaction of the individual, the environment, and the task being attempted.”
- “Therapists must consider how the characteristics, resources, and constraints of each influence the ability to move, and how each might be manipulated during intervention to enhance motor learning and effectiveness of movement.”





Theories of motor control



Reflex Theory

- Sir Charles Sherrington in 1906
- Reflexes are the basis of complex behaviour
- Stimulus proves a reflex which becomes the stimulus for the next response

- Implications: Bobath concept

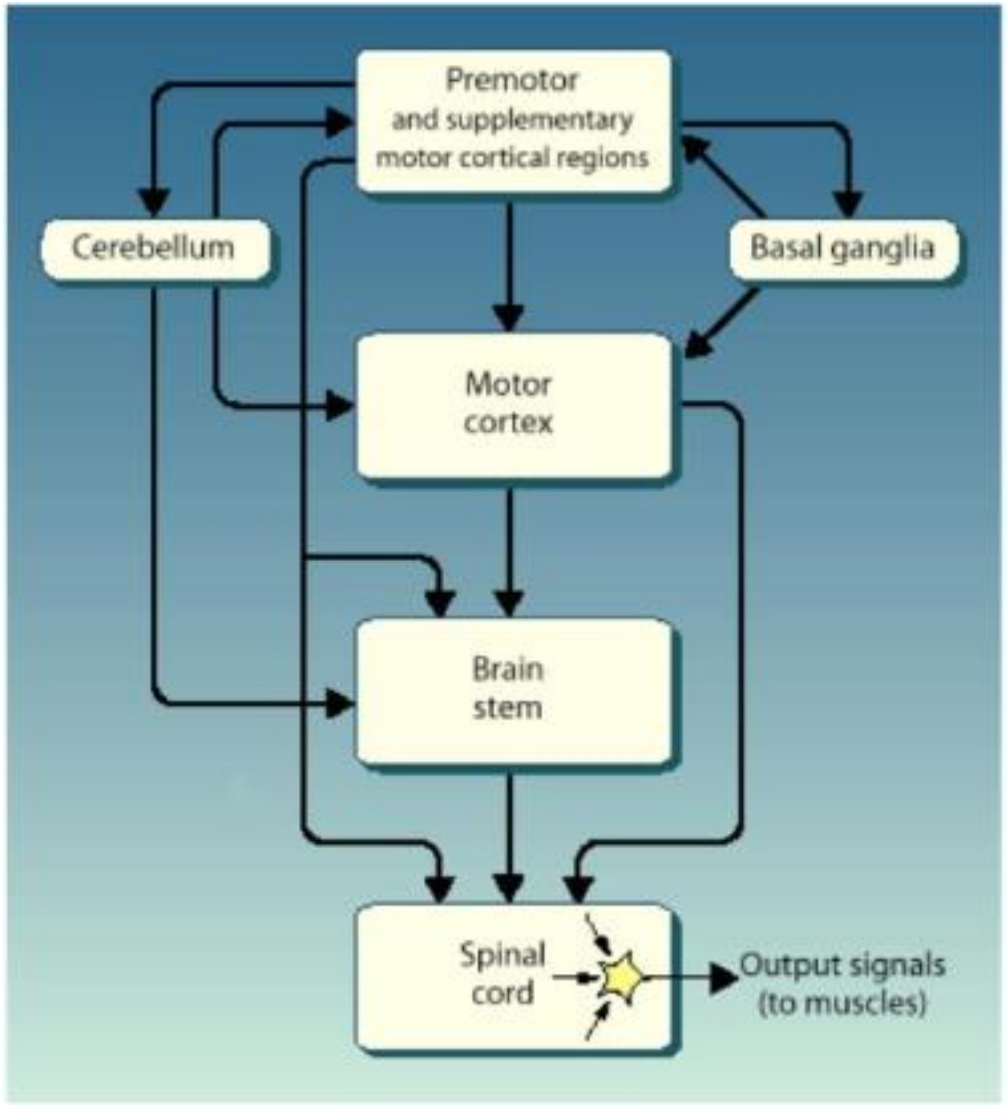
- Limitations: Does not explain:
  - voluntary movement
  - how a single stimulus may elicit a variety of responses depending on context

Hierarchical Theory (Adams 1971)

- Hierarchical organization of the CNS
- Cortical centres control movement in a top-down manner throughout the nervous system.
- Closed loop mode: sensory feedback is needed and used to control movement
- Voluntary movement is initiated by “will”

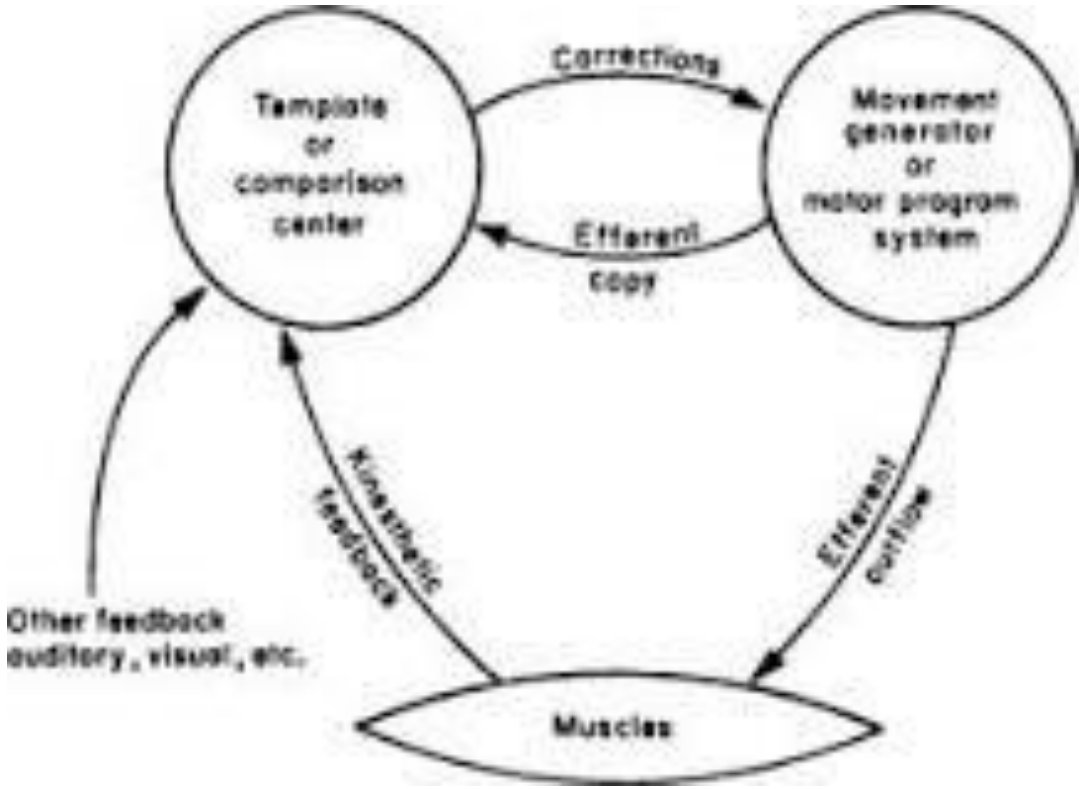
- **Implications:** identify and prevent primitive reflexes, normalise tone, developmental sequencing, restore normal patterns

- **Limitations:** does not explain how a reflex on the lowest hierarchical level would dominate motor function (withdrawal reflex).



Motor Programming theory – Schmidt 1976)

- Focused on the nature of action rather than the response
- Motor response can be a response to a stimulus or by a central process
- Central motor pattern – capable of generating movement in the absence of a stimulus
- **Implications:**
  - Abnormal movements – not just reflexes
  - Motor relearning required for functional tasks
  - Don't just re-educate muscles in isolation
- **Limitations**
  - Consider the role of musculoskeletal and environmental variables to achieve movement control



Systems Theory (Bernstein 1967)

- ‘‘movements are not controlled centrally nor peripherally, but rather are effected by interactions among multiple systems’’
- Recognises that the body is a mechanical system that responds differently to the same command due to the interactions between external forces.

- **Implications:**
  - Assessment and treatment should examine deficits in specific systems contributing to motor control e.g. MSK, perceptual etc

- **Limitations:**
  - Systems theory does not account for the subject’s interaction with the environment.



Other theories

- Ecological Theory (Gibson 1960)
- Dynamic Action Theory (multiple)
- Theory of parallel distributed processing
- Action oriented theory

- Explore these and find the implications for the planning of assessment and treatment of patients with neurological problems.





Any questions

Additional Reading

- Cano-de-la-Cuerda R et al., (2015). Theories and control models and motor learning: Clinical applications in neurorehabilitation. Neurología. 2015;30(1):32—41

- Shumway Cook A, et al. (2022). Motor Control: Translating Research into Clinical Practice. Motor Control: Issues and Theories: Chapter 1. Wolters Kluver, Lippincott Williams & Wilkins